

The slide features a solid blue background. On the left and right edges, there are decorative patterns of overlapping chevron shapes in yellow, magenta, and light blue. The main text is centered in the upper half of the slide.

Advanced SIP Routing

with FreeSWITCH modules

SIP Routing - Headers

VIA headers

Route headers



SIP Routing - VIA Headers



When a UAC sends a request it adds a VIA header to the message

Each proxy adds a VIA header before passing the request to the next node



SIP Routing - Route Headers

Route headers force routing thru a proxy

Future requests follow Route headers

SIP Routing - Strict Routing



R-URI rewritten (from the top-most Route header) to contain the address of the next hop

Original R-URI saved as the last Record Route header



SIP Routing - Loose Routing

R-URI is never re-written, it always contains the address of the destination

The next hop is the address contained in the top-most Route header

SIP Routing - Recap

Requests (subsequent) follow Route headers

Responses follow VIA headers

FreeSWITCH Modules

160 Modules

Routing Modules

Distributor (mod_distributor)
Easy Route (mod_easyroute)
Nibble Bill (mod_nibblebill)
Least Cost Routing (mod_lcr)

Distributor (mod_distributor)

Round-robin routing to gateways

Configuration via a file, not DB

Multiple lists of gateways can be defined

Easy Route (mod_easyroute)

Simple inbound DID routing engine

Configuration via DB, not a file

Provides number translation and setting of an account code

Nibble Bill (mod_nibblebill)

Realtime pre-paid / post-paid / pay-per-call

Configuration via DB, not file

Bill at different rates during a call

Nibble Bill (mod_nibblebill)

Low balance warnings

Disconnect or re-route a call when the balance falls to zero

Maximum credit / Fraud detection

Least Cost Routing (mod_lcr)

Route calls via user defined criteria:
cost, reliability, quality, ...

Supports multiple profiles

Profiles allow for unique rules for
different customer types.

Least Cost Routing (mod_lcr)

Profiles can be used to modify how data is returned from the DB

Can be integrated with Limit

Supports (longest) prefix matching

Longest Prefix Matching



Algorithm used by routers to select a routing table entry

Adapted to the telecom world to select a (least cost) routing table entry



Least Cost Routing (mod_lcr)

Can be integrated with Nibble Bill

Supports intralata / intrastate routing

Can override CallerID format per carrier

Least Cost Routing (mod_lcr)

carriers table - defines the carriers

carrier_gateway table - defines gateway information for each carrier

lcr table - defines rules for a given digit sequence

Least Cost Routing (mod_lcr)

Transferred to from the dialplan

Bridged (save results and bridge)

Command Line or ESL

Extremely flexible

Thank you

Questions ? Comments ?

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